

repositories. For example:

```
> new.packages()
[1] "A3" "aaSEA"
[3] "AATtools" "aba"
[5] "ABACUS" "abbreviate"
[7] "abbyyR" "abc"
[9] "abc.data" "ABC.RAP"
[11] "abcADM" "ABCanalysis"
[13] "abcdeFBA" "ABCOptim"
[15] "ABCP2" "abcrf"
[17] "abcrlda" "abctools"
...
```

You can download the R packages to a specific directory using the `download.packages` function.

The *surveillance* R package provides statistical methods for modelling and monitoring of time series data and for epidemic phenomena. Its R source code can be downloaded to a `'tmp'` directory, as illustrated below:

```
> download.packages("surveillance", destdir="/tmp")
trying URL 'https://cloud.r-project.org/src/contrib/surveillance_1.19.1.tar.gz'
Content type 'application/x-gzip' length 4388559 bytes (4.2 MB)
=====
downloaded 4.2 MB
[1] [2]
[1,] "surveillance" "/tmp/surveillance_1.19.1.tar.gz"
```

Layout

The *bitops* R package implements bitwise operations on integer vectors. The *bitops.1.0.7* source code contains the following files:

```
$ ls
ChangeLog DESCRIPTION INDEX man MD5 NAMESPACE R
README.md src tests
$ ls man
bitAnd.Rd bitFlip.Rd bitShiftL.Rd cksum.Rd
$ ls R
bitops.R
$ ls src/
bit-ops.c bit-ops.h cksum.c init.c
$ ls tests/
consistency.R
```

An R source code package should contain the following files.

- **README:** This documentation file contains the purpose of the project, and the steps to install, configure and use the package. For example, the README of the

bitops.1.0.7 project contains the following text:

```
$ cat README.md

# R-bitops -- R CRAN package 'bitops' -- implementing bitwise
operations
## Functionality
Bitwise operations on (the 32-bit) \R integers
## Context and History
Probably the smallest [CRAN](https://CRAN.R-project.org)
[R](https://www.r-project.org) package I maintain,
the package has a long history, originating as S / S-plus
extensions by
Steve Dutky.
...
```

- **ChangeLog:** This file provides a brief history of the changes for the various commits to the project. The *ChangeLog* file in *bitops.1.0.7* shows the recent changes on the top of the file.

```
$ cat ChangeLog

2021-04-13 Martin Maechler <maechler@stat.math.ethz.ch>
* From Dan Robertson (dlobertson) via GH PR #3,
Feb.2016:
    Introduce C-style "operators":
    "%&%", "%|%", "%^%", "%<%", "%>%"
2021-03-30 Martin Maechler <maechler@stat.math.ethz.ch>
* src/bit-ops.c (R_2_UINT_): cast even more, now via
  _2_UINT_()
2021-03-24 Martin Maechler <maechler@stat.math.ethz.ch>
* DESCRIPTION (Date, URL): cosmetic before finally
releasing
...
```

- **INDEX:** This file is optional and provides names of important objects in the package, along with their description. For example:

```
$ cat INDEX

bitAnd          Bitwise And, Or and Xor operations
bitFlip         Binary Flip (Not) Operator
bitShiftL       Bitwise Shift Operator (to the Left
or Right)
cksum           Compute Check Sum
```

- **MD5:** This file contains the MD5 output of the source files in the R package. This is used to verify the checksum of the downloaded files using the *md5sum* utility.
- **NAMESPACE:** This file contains directives that specify the various imports and exports of the name